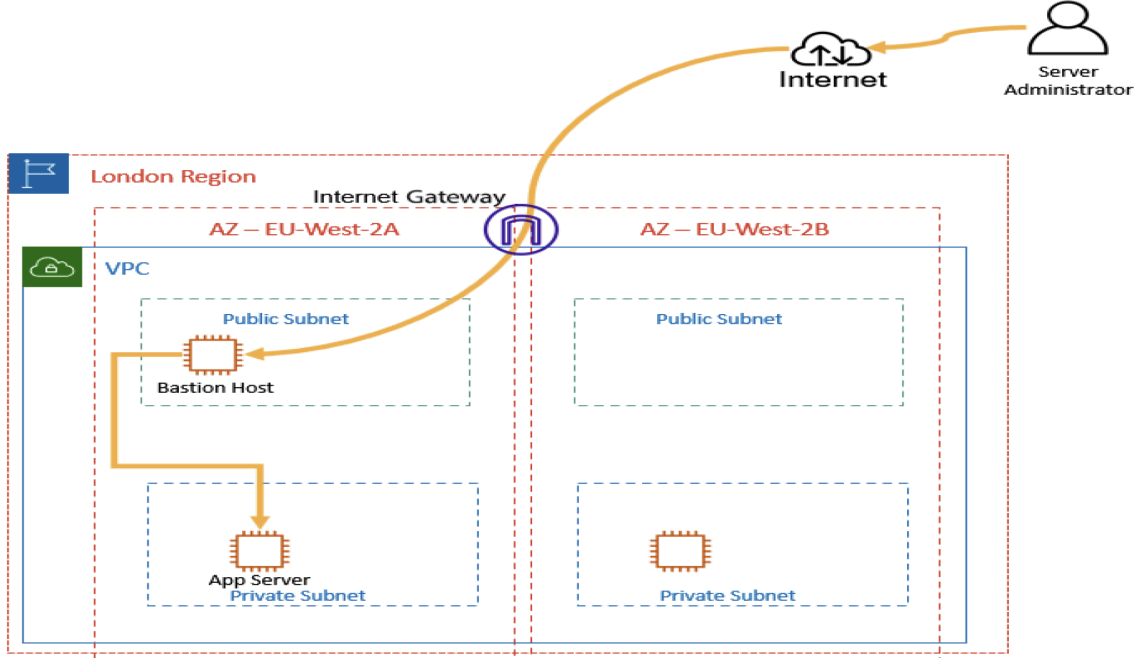




Lab: Custom VPC + Bastion Host + Apache Web Server



Hedef

AWS üzerinde kullanıcıya özel bir VPC kurmak; Public ve Private subnet'ler, Internet Gateway (IGW), NAT Gateway, Bastion Host ve bir Apache web sunucusu oluşturup erişim ve yönlendirme testlerini gerçekleştirmek.



Ana Adımlar

1. **VPC ve Subnet'ler oluştur** → Public + Private subnet
2. **Internet Gateway & NAT Gateway ekle** → Public internete çıksın, Private NAT üzerinden çıksın
3. **Route Tables yapılandır** → Public → IGW, Private → NAT
4. **Security Groups oluştur** → Bastion için SSH, Web için HTTP/HTTPS, DB için sınırlı erişim
5. **Bastion Host kur (Public Subnet)** → Private subnet'e köprü
6. **Web Server kur (Public Subnet)** → Apache yükle, test et
7. **Opsiyonel Private EC2** → Bastion üzerinden erişim doğrula
8. **Erişim Testleri** (IGW, NAT, Bastion, routing, SG, NACL)
9. **Cleanup** (kaynakları güvenli silme)

Ayrıntılı Lab Adımları

1 VPC ve Subnet'ler

- AWS Console → VPC → “Create VPC”
 - Vpc Settings: VPC only
 - Name tag: Workshop-VPC
 - IPv4 CIDR block: 10.0.0.0/16

VPC > Your VPCs > Create VPC

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

Workshop-VPC

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	Workshop-VPC	Remove tag
Add tag		

You can add 49 more tags

[Cancel](#) [Preview code](#)

Subnet oluşturma:

- Public Subnet

Subnets→Create Subnet

- **VPC ID:** Workshop-VPC
- **Subnet Name:** Public Subnet A
- **Availability Zone:** us-east-1a
- **IPv4 subnet CIDR block:** 10.0.1.0/24

Create subnet [Info](#)

VPC
VPC ID
vpc-04e12e32d9a3f68dd (Workshop-VPC)
Associated VPC CIDRs
IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public Subnet A
The name can be up to 250 characters long.
Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az2 (us-east-1a)
IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.1.0/24
256 IPs
Tags - optional

Key	Value - optional	
Q Name	Public Subnet A	Remove

[Add new tag](#)
You can add 49 more tags.
[Remove](#)
[Add new subnet](#)

[Cancel](#) [Create subnet](#)

- Private Subnet

Subnets→Create Subnet

- **VPC ID:** Workshop-VPC
- **Subnet Name:** Private Subnet A
- **Availability Zone:** us-east-1b
- **IPv4 subnet CIDR block:** 10.0.2.0/24

Create subnet Info

VPC
VPC ID
Create subnets in this VPC.
vpc-04e12e32d9a3f68dd (Workshop-VPC)
Associated VPC CIDRs
IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1
Subnet name
Create a name within the length of 3 to 255 characters and a value that you specify.
Private Subnet A
The name can be up to 255 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az4 (us-east-1b)
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.2.0/24 256 IPs

Tags - optional

Key	Value - optional	
Q Name	Q Private Subnet A	Remove

[Add new tag](#)
You can add 49 more tags.
[Remove](#)
[Add new subnet](#)

[Cancel](#) [Create subnet](#)

2) Internet Gateway (IGW) ve NAT Gateway

Internet gateway:

VPC→Internet gateways→Create internet gateway

Internet gateway settings:

- Name: Workshop-IGW

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Create a tag with a key of `Name` and a value that you specify.

Workshop-IGW

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q Workshop-IGW X Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway

Internet gateway VPC'ye attach etme

internet gateway seç Actions dan Attach to VPC seç

The following internet gateway was created: igw-0a71c2c7d33fdb41e - Workshop-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet.

Internet gateways (1/2) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
Workshop-IGW	igw-0a71c2c7d33fdb41e	detached	--	193801312183

Actions

- Attach to a VPC
- View details
- Attach to VPC
- Detach from VPC
- Manage tags
- Delete internet gateway

VPC içerisinde **Workshop-VPC** seç ve **Attach internet gateway** butonuna bas.

The following internet gateway was created: igw-0a71c2c7d33fdb41e - Workshop-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to VPC (igw-0a71c2c7d33fdb41e) Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC

Q vpc-04e12e32d9a3f68dd X

AWS Command Line Interface command

Cancel Attach internet gateway

Internet gateways (1/2) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
Workshop-IGW	igw-0a71c2c7d33fdb41e	Attached	vpc-04e12e32d9a3f68dd Workshop-VPC	193801312183

- NAT Gateway oluşturma:

VPC→NAT gateways→Create NAT gateway

Internet gateway settings:

- **Name:** Workshop-NAT
- **Subnet:** Public Subnet A
- **Connectivity type:** Public
- **Elastic IP allocation ID:** Allocate Elastic IP

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Subnet
Select a subnet in which to create the NAT gateway.

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.
 [Allocate Elastic IP](#)

► **Additional settings** [Info](#)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

[Remove](#)

[Add new tag](#)

You can add 49 more tags.

[Cancel](#) [Create NAT gateway](#)

✔ NAT gateway nat-0ed11662a0752325b | Workshop-NAT was created successfully.

NAT gateways (1) [Info](#)

[Actions](#) [Create NAT gateway](#)

Name	NAT gateway ID	Connectivity...	State	State message	Primary public I...	Primary private I...	Primary network...	VPC
Workshop-NAT	nat-0ed11662a0752325b	Public	Pending	-	-	10.0.1.56	eni-0ff03308077f12...	vpc-f

NAT gateway durumu 1 kaç dakika içinde aktif olacaktır

- Bu sayede:

- Public Subnet → IGW üzerinden dışarı
- Private Subnet → NAT üzerinden dışarı

3 Route Tables

- Public Route Table:

VPC→Route tables→Create route table

- Name: Pubic-RT
- VPC: Workshop-VPC

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

You can add 49 more tags.

Route table rtb-00189afd105e6e210 | Pubic-RT was created successfully.

Route tables (1/3) Info

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-01e2c572623d01b16	-	-	Yes	vpc-0894201adb1a5b443	193801312183
☐	-	rtb-0f5e491d174b640c2	-	-	Yes	vpc-04e12e32d9a3f68dd Wor...	193801312183
☒	Pubic-RT	rtb-00189afd105e6e210	-	-	No	vpc-04e12e32d9a3f68dd Wor...	193801312183

Route Ekle

Public-RT→Routes→Edit routes→Add route

- Destination: 0.0.0.0/0
- Target: internet Gateway(Workshop-IGW)

Edit routes

Destination
10.0.0.0/16

Target

Status
Active

Propagated
No

Route Origin
CreateRouteTable

Route Table'ı Public Subnet ile ilişkilendir

Public-RT→Subnet associations→Edit subnet associations→Select Public subnet A

- Available subnets: Public subnet A

VPC > Route tables > rtb-00189afd105e6e210 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public Subnet A	subnet-0a068da1cd66d67b5	10.0.1.0/24	-	Main (rtb-0f5e491d174b640c)
☐	Private Subnet A	subnet-0a584060c8c9925b5	10.0.2.0/24	-	Main (rtb-0f5e491d174b640c)

Selected subnets

subnet-0a068da1cd66d67b5 / Public Subnet A

Cancel Save associations

- Private Route Table:

VPC→Route tables→Create route table

- Name: Private-RT
- VPC: Workshop-VPC

VPC > Route tables > Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Private-RT

VPC

vpc-04e12e32d9a3f68dd (Workshop-VPC)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Private-RT

Add new tag

You can add 49 more tags.

Cancel Create route table

Route table rtb-0c8f27ee7ca2f0ecb | Private-RT was created successfully.

Route tables (1/4) info

Last updated 1 minute ago

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-01e2c572623d01b16	-	-	Yes	vpc-0894201adb1a5b443	193801312183
-	rtb-0f5e491d174b640c2	-	-	Yes	vpc-04e12e32d9a3f68dd Wor...	193801312183
Public-RT	rtb-00189afd105e6e210	subnet-0a068da1cd66d6...	-	No	vpc-04e12e32d9a3f68dd Wor...	193801312183
Private-RT	rtb-0c8f27ee7ca2f0ecb	-	-	No	vpc-04e12e32d9a3f68dd Wor...	193801312183

rtb-0c8f27ee7ca2f0ecb / Private-RT

Details **Routes** Subnet associations Edge associations Route propagation Tags

Routes (1)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

Route Ekle

Private-RT→Routes→Edit routes→Add route

- Destination: 0.0.0.0/0
- Target: NAT Gateway(Workshop-NAT)

VPC > Route tables > rtb-0c8f27ee7ca2f0ecb > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway		No	CreateRoute

Add route

Cancel Preview **Save changes**

Route Table'ı Private Subnet ile ilişkilendir

Private-RT→Subnet associations→Edit subnet associations→Select private subnet A

- Available subnets: Private subnet A

VPC > Route tables > rtb-0c8f27ee7ca2f0ecb > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public Subnet A	subnet-0a068da1cd66d67b5	10.0.1.0/24	-	rtb-00189afd105e6e210 / Pul
Private Subnet A	subnet-0a584060c8c9925b5	10.0.2.0/24	-	Main (rtb-0f5e491d174b640c

Selected subnets

subnet-0a584060c8c9925b5 / Private Subnet A

Cancel **Save associations**

4 Security Groups

VPC→Security→Security groups→Create Security Group

- **SG-Bastion**
 - **Security group name:** SG-Bastion
 - **Description:** SG-Bastion
 - **VPC:** Workshop-VPC
 - **Inbound rules:**
 - **Type:** SSH
 - **Protocol:** TCP
 - **Port range:** 22
 - **Source:** 0.0.0.0/0

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

SG-Bastion

Name cannot be changed after creation.

Description Info

SG-Bastion

VPC Info

vpc-04e12e32d9a3f68dd (Workshop-VPC)

Inbound rules Info

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	Anywhere... 0.0.0.0/0	

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules Info

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom 0.0.0.0/0	

[Add rule](#)

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags

[Cancel](#) [Create security group](#)

- **SG-Web**

- **Security group name:** SG-Web
- **Description:** SG-Web
- **VPC:** Workshop-VPC
- Inbound: 80/tcp → 0.0.0.0/0
- Inbound: 22/tcp → SG-Bastion

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

SG-Web

Name cannot be edited after creation.

Description [Info](#)

SG-Web

VPC [Info](#)

vpc-04e12e32d9a3f68dd (Workshop-VPC)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	Custom sg-035ad96a4d168e4bb	
HTTP	TCP	80	Anywhere... 0.0.0.0/0	

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom 0.0.0.0/0	

[Add rule](#)

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Create security group](#)

5 Bastion Host (Public Subnet)

EC2→Instances→Launch an instance

- **Name and tags:** Bastion
- **AMI**→ Amazon Linux 2023, t3.micro

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Bastion

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0b09ffb6d8b58ca91 (64-bit (x86), uefi-preferred) / ami-0b2f2a5d7a4d7f37f (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250908.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86)	uefi-preferred	ami-0b09ffb6d8b58ca91	2025-09-06	ec2-user	Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- **Key Pair (login):**
 - **Create new key pair** → **Key pair name:** ec2_ssh_key
 - **Type:** RSA, **File format:** .pem → **Create ve seç.**

Create key pair

×

Key pair name

Key pairs allow you to connect to your instance securely.

ec2_ssh_key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel

Create key pair

- Network settings → Edit:
 - VPC: Workshop-VPC
 - Subnet: Publicsubnet A
 - Auto-assign public IP: Enable
 - Firewall (SG): Select security group
 - Security group name: SG-Bastion

▼ Network settings

Info

VPC - required

Info

vpc-04e12e32d9a3f68dd (Workshop-VPC)

10.0.0/16

▼

↻

Subnet

Info

subnet-0a068da1cd66d67b5

Public Subnet A

VPC: vpc-04e12e32d9a3f68dd

Owner: 193801312183

Availability Zone: us-east-1a (use1-az2)

Zone type: Availability Zone

IP addresses available: 250

CIDR: 10.0.1.0/24

▼

↻

Create new subnet

Auto-assign public IP

Info

Enable

▼

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups

Info

Select security groups

▼

SG-Bastion sg-035ad96a4d168e4bb

VPC: vpc-04e12e32d9a3f68dd

×

↻

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Launch instance butonuna bas sunucu running olunca:

Bastion i-0d6b5f15295f1004d Running t2.micro Initializing View alarms + us-east-1a

i-0d6b5f15295f1004d (Bastion)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID i-0d6b5f15295f1004d	Public IPv4 address 44.212.21.106 open address	Private IPv4 addresses 10.0.1.138
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-138.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-1-138.ec2.internal	

SSH bağlantısı test et:

```
ssh -i ec2_ssh_key.pem ec2-user@<Bastion_PublicIP>
```

```
chmod 400 ec2_ssh_key.pem
ssh -i ec2_ssh_key.pem ec2-user@44.212.21.106
```

```
(base) Downloads ➡ chmod 400 ec2_ssh_key.pem
(base) Downloads ➡ ssh -i ec2_ssh_key.pem ec2-user@44.212.21.106
The authenticity of host '44.212.21.106 (44.212.21.106)' can't be established.
ED25519 key fingerprint is SHA256:PJl5G7mlceJLgz1lVeLTHzAc2dAR5/mDI7pFbh00FhA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.212.21.106' (ED25519) to the list of known hosts.
#_
~\_ #####_ Amazon Linux 2023
~~ \_#####\
~~ \###|
~~ \#/ ___ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~ /
~~._. _/
~~ _/ _/
~~ _/m/'
[ec2-user@ip-10-0-1-138 ~]$
```

6 Web Server (Public Subnet)

EC2→Instances→Launch an instance

- Name and tags: Web
- AMI→ Amazon Linux 2023, t3.micro

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Web

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

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Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-0b09ffb6d8b58ca91 (64-bit (x86), uefi-preferred) / ami-0b2f2a5d7a4d7f37f (64-bit (Arm), uefi-preferred)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250908.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86)	uefi-preferred	ami-0b09ffb6d8b58ca91	2025-09-06	ec2-user	Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

- Key Pair (login): Select Key pair: ec2_ssh_key

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

ec2_ssh_key

[Create new key pair](#)

- **Network settings** → **Edit**:
 - **VPC**: Workshop-VPC
 - **Subnet**: Public subnet A
 - **Auto-assign public IP**: Enable
 - **Firewall (SG)**: Select security group
 - **Security group name**: SG-Web

▼ **Network settings** [Info](#)

VPC - required [Info](#)

vpc-04e12e32d9a3f68dd (Workshop-VPC)
10.0.0.0/16

Subnet [Info](#)

subnet-0a068da1cd66d67b5 Public Subnet A
VPC: vpc-04e12e32d9a3f68dd Owner: 193801312183 Availability Zone: us-east-1a (use1-az2)
Zone type: Availability Zone IP addresses available: 249 CIDR: 10.0.1.0/24

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

SG-Web sg-0fe3e9238a5c97845
VPC: vpc-04e12e32d9a3f68dd

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► **Advanced network configuration**

- **Advanced details**:
 - **User data**: aşağıdaki script'i yapıştır.

```
#!/bin/bash
sudo dnf update -y
sudo dnf install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<html><h1> Welcome to Tech istanbul: Apache Web Server</h1><html>" >
/var/www/html/index.html
```


User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
sudo dnf update -y
sudo dnf install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<html><h1> Welcome to Tech istanbul: Apache Web Server</h1><html>" >
/var/www/html/index.html
```

Launch instance butonuna bas sunucu running olunca:

The screenshot shows the AWS Management Console. At the top, there's a search bar and a dropdown for 'All states'. Below is a table of EC2 instances. The 'Web' instance is selected and highlighted with a red box. The table has columns: Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, Public IPv4 DNS, Public IPv4 address, Elastic IP, IPv6 IPs, Monitoring, Security group name, and Key name. The 'Web' instance is in the 'Running' state, type 't2.micro', and has a public IPv4 address of 3.82.122.163. Below the table, the 'Details' tab is selected for the 'Web' instance. The 'Instance summary' section shows the instance ID, name, and type. The 'Public IPv4 address' is highlighted with a red box and shows the address 3.82.122.163 with a link to 'open address'. The 'Instance state' is 'Running'. The 'Private IP DNS name (IPv4 only)' is 'ip-10-0-1-15.ec2.internal'. The 'Instance type' is 't2.micro'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 address	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name
Web	i-0e1376d0a22edad10	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	3.82.122.163	-	-	disabled	SG-Web	ec2_ssh
Bastion	i-006b5f15295f1004d	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	44.212.21.106	-	-	disabled	SG-Bastion	ec2_ssh

i-0e1376d0a22edad10 (Web)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary

Instance ID: i-0e1376d0a22edad10

IPv4 address: -

Hostname type: IP name: ip-10-0-1-15.ec2.internal

Answer private resource DNS name: -

Public IPv4 address: 3.82.122.163 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-10-0-1-15.ec2.internal

Instance type: t2.micro

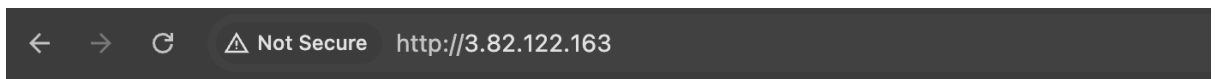
Private IPv4 addresses: 10.0.1.15

Public DNS: -

Elastic IP addresses: -

Web bağlantısı test et:

- Tarayıcıdan → [http://<Web_EC2_Public_IP>](http://3.82.122.163)



Welcome to Tech istanbul: Apache Web Server

SSH bağlantısı test et:

`ssh -i ec2_ssh_key.pem ec2-user@<Web Server PublicIP>`

```
ssh -i ec2_ssh_key.pem ec2-user@3.82.122.163
```

Web Server SSH bağlantısı olmaz çünkü Security group 22 port sadece Bastion sunucuya izinli

```
(base) Downloads ➡ ssh -i ec2_ssh_key.pem ec2-user@3.82.122.163
kex_exchange_identification: read: Connection reset by peer
Connection reset by 3.82.122.163 port 22
(base) Downloads ➡
```

Web Server'a SSH ile bağlanmak için Bastion Host sunucusuna kullanacağız.

```
ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p
ec2-user@44.212.21.106' \
ec2-user@10.0.1.15
```

```
(base) Downloads ➡ ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p ec2-user@44.212.21.106' \
ec2-user@10.0.1.15
The authenticity of host '10.0.1.15 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:KI5CkYhfw+5/ts0If6a13Esz9Q0UR1NF73gjn1C3kZs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.1.15' (ED25519) to the list of known hosts.
,      #_
~\_   #####_      Amazon Linux 2023
~~  \#####\
~~   \###|
~~    \#/  ___  https://aws.amazon.com/linux/amazon-linux-2023
~~      V~'  '->
~~~~
~~~_._._/_/_/
~~/_/_/
~~/_m/'
ec2-user@ip-10-0-1-15 ~]$
```

7 Private EC2

EC2→Instances→Launch an instance

- **Name and tags:** `private-EC2`
- **AMI**→ `Amazon Linux 2023, t3.micro`

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

private-EC2

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

Free tier eligible

ami-0b09ffb6d8b58ca91 (64-bit (x86), uefi-preferred) / ami-0b2f2a5d7a4d7f37f (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250908.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-0b09ffb6d8b58ca91	2025-09-06	ec2-user

[Verified provider](#)

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software.](#)

- **Key Pair (login):** Select Key pair: `ec2_ssh_key`

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

ec2_ssh_key

[Create new key pair](#)

- **Network settings** → **Edit**:
 - **VPC**: Workshop-VPC
 - **Subnet**: Private subnet A
 - **Auto-assign public IP**: disable
 - **Firewall (SG)**: Select security group
 - **Security group name**: SG-private
 - **Inbound**: 22/tcp → SG-Bastion

▼ Network settings

Info

VPC - required

Info

vpc-04e12e32d9a3f68dd (Workshop-VPC)

10.0.0.0/16

↻

Subnet

Info

subnet-0a584060c8c9925b5

Private Subnet A

↻

Create new subnet

VPC: vpc-04e12e32d9a3f68dd

Owner: 193801312183

Availability Zone: us-east-1b (use1-az4)

Auto-assign public IP

Info

Disable

▼

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

SG-private

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/[]+=&()!\$*

Description - required

Info

SG-private

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, sg-035ad96a4d168e4bb)

Remove

Type

Info

ssh

▼

Protocol

Info

TCP

Port range

Info

22

Source type

Info

Custom

▼

Source

Info

Q Add CIDR, prefix list or security group

sg-035ad96a4d168e4bb

×

Description - optional

Info

e.g. SSH for admin desktop

Add security group rule

► Advanced network configuration

Launch instance butonuna bas sunucu running olunca:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name	Launch time
private-EC2	i-09afc2c714dd4d864	Running	t2.micro	Initializing	View alarms +	us-east-1b	-	-	-	-	disabled	SG-private	ec2_ssh_key	2025/09/13 14:21 GMT+3
	i-0d4c19901d0a0644	Terminated	t2.micro	-	View alarms +	us-east-1b	-	-	-	-	disabled	-	ec2_ssh_key	2025/09/13 13:39 GMT+3
Web	i-0a151760a22ed0e10	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	3.82.122.163	-	-	disabled	SG-Web	ec2_ssh_key	2025/09/13 13:57 GMT+3
Bastion	i-0d05f15295f10d4f	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	44.212.21.106	-	-	disabled	SG-Bastion	ec2_ssh_key	2025/09/13 13:47 GMT+3

i-09afc2c714dd4d864 (private-EC2)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID i-09afc2c714dd4d864

IPv6 address -

Hostname type IP name ip-10-0-2-190.ec2.internal

Answer private resource DNS name -

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-2-190.ec2.internal

Instance type

t2.micro

Private IPv4 addresses

10.0.2.190

Public DNS

-

Elastic IP addresses

-

SSH bağlantısı test et:

Private-EC2'ya SSH ile bağlanmak için Bastion Host sunucusuna kullanacağız.

BASTION_PUBLIC_IP: 44.212.21.106

Private-EC2_IP: 10.0.2.190

```
ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p
ec2-user@44.212.21.106' \
ec2-user@10.0.2.190
```

```
base) Downloads ➡ ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p ec2-user@44.212.21.106' \
ec2-user@10.0.2.190
The authenticity of host '10.0.2.190 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:RYQxVturE4szlqhcUhM3+dS5l4fDajZNWTl18BYTZ/Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.190' (ED25519) to the list of known hosts.
#_
~\_ #####_ Amazon Linux 2023
~~\_#####\
~~\_###|
~~\_#/____ https://aws.amazon.com/linux/amazon-linux-2023
~~_V~' '->
~~~_/_/
~~~_/_/_/
~~~_/_/_'
ec2-user@ip-10-0-2-190 ~]$
```

```
[ec2-user@ip-10-0-2-190 ~]$ sudo yum install nginx
Amazon Linux 2023 Kernel Livepatch repository                209 kB/s | 21 kB    00:00
Dependencies resolved.

=====
Package                Architecture Version                Repository            Size
=====
Installing:
nginx                  x86_64          1:1.28.0-1.amzn2023.0.2  amazonlinux           33 k
Installing dependencies:
generic-logos-httpd    noarch          18.0.0-12.amzn2023.0.3  amazonlinux           19 k
gperftools-libs        x86_64          2.9.1-1.amzn2023.0.3    amazonlinux           308 k
libunwind              x86_64          1.4.0-5.amzn2023.0.2    amazonlinux           66 k
nginx-core             x86_64          1:1.28.0-1.amzn2023.0.2  amazonlinux           686 k
nginx-filessystem       noarch          1:1.28.0-1.amzn2023.0.2  amazonlinux           9.6 k
nginx-mimetypes        noarch          2.1.49-3.amzn2023.0.3    amazonlinux           21 k
=====

Transaction Summary
=====
Install 7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Is this ok [y/N]:
```

Nat Gateway Testi:

Private Subnet route table **0.0.0.0/0** networkü için NAT gateway'e yönlendirildiği için. NAT gateway sayesinde Private-EC2 sunucusunun ihtiyacı olan tüm paketler internetten sunucumuza yüklenebilir. Şimdi PrivateRT tablosundan **0.0.0.0/0** → NAT gateway yönlendirmesini kaldırıp tekrar httpd paketini yüklemeyi deneyelim.

VPC→Route tables→PrivateRT→Edit routes
0.0.0.0/0 yönlendirmesini remove et ve save changes butonuna bas

The screenshot shows the AWS Management Console interface for editing routes in a Private Route Table. The table has two columns: 'Destination' and 'Target'. The first row shows a route for destination '10.0.0.0/16' with target 'local'. The second row shows a route for destination '0.0.0.0/0' with target 'NAT Gateway'. The 'NAT Gateway' route is highlighted with a red box. A 'Remove' button is visible next to it. At the bottom, there are 'Cancel', 'Preview', and 'Save changes' buttons. The 'Save changes' button is highlighted with a red box.

```
[ec2-user@ip-10-0-2-190 ~]$ sudo yum install httpd
Last metadata expiration check: 0:07:14 ago on Sat Sep 13 11:25:53 2025.
Dependencies resolved.
=====
Package                Architecture    Version                Repository             Size
=====
Installing:
httpd                  x86_64         2.4.64-1.amzn2023.0.1  amazonlinux            47 k
Installing dependencies:
apr                    x86_64         1.7.5-1.amzn2023.0.4   amazonlinux            129 k
apr-util               x86_64         1.6.3-1.amzn2023.0.1   amazonlinux            98 k
httpd-core             x86_64         2.4.64-1.amzn2023.0.1  amazonlinux            1.4 M
httpd-filesystem       noarch         2.4.64-1.amzn2023.0.1  amazonlinux            13 k
httpd-tools            x86_64         2.4.64-1.amzn2023.0.1  amazonlinux            81 k
libbrotli              x86_64         1.0.9-4.amzn2023.0.2   amazonlinux            315 k
mailcap                noarch         2.1.49-3.amzn2023.0.3  amazonlinux            33 k
Installing weak dependencies:
apr-util-openssl       x86_64         1.6.3-1.amzn2023.0.1   amazonlinux            17 k
mod_http2              x86_64         2.0.27-1.amzn2023.0.3  amazonlinux            166 k
mod_lua                x86_64         2.4.64-1.amzn2023.0.1  amazonlinux            60 k

Transaction Summary
=====
Install 11 Packages

Total download size: 2.3 M
Installed size: 6.9 M
Is this ok [y/N]: y
Downloading Packages:
[=====] --- B/s | 0 B --:-- ETA
```

Paketin bir türlü indirilmediğini gördük. Çünkü sunucumuz artık internete çıkamıyor.

Cleanup (Lab Sonunda)

1. EC2 instance'larını terminate et
2. Elastic IP release et
3. NAT Gateway sil
4. IGW detach & delete
5. Subnet & Route Tables sil
6. VPC sil